

Apêndice

Propriedades Gerais



TABELA A.1

Fatores de Conversão

Área (*A*)

$$\begin{aligned} 1 \text{ mm}^2 &= 1,0 \times 10^{-6} \text{ m}^2 & 1 \text{ ft}^2 &= 144 \text{ in}^2 \\ 1 \text{ m}^2 &= 10,7669 \text{ ft}^2 & 1 \text{ ft}^2 &= 0,092903 \text{ m}^2 \end{aligned}$$

Capacidade térmica (C_p , C_v , C), entropia específica (s)

$$1 \text{ kJ}/(\text{kg} \times \text{K}) = 0,238846 \text{ Btu}/(\text{lbm} \times \text{R}) \quad 1 \text{ Btu}/(\text{lbm} \times \text{R}) = 4,1868 \text{ kJ}/(\text{kg} \times \text{K})$$

Coefficiente de transferência de calor (h)

$$1 \text{ W}/(\text{m}^2 \times \text{K}) = 0,17611 \text{ Btu}/(\text{h} \times \text{ft}^2 \times \text{R}) \quad 1 \text{ Btu}/(\text{h} \times \text{ft}^2 \times \text{R}) = 5,67826 \text{ W}/(\text{m}^2 \times \text{K})$$

Comprimento (L)

$$\begin{aligned} 1 \text{ mm} &= 0,001 \text{ m} = 0,1 \text{ cm} & 1 \text{ in} &= 0,0254 \text{ m} \\ 1 \text{ m} &= 3,28084 \text{ ft} = 39,370 \text{ in} & 1 \text{ ft} &= 12 \text{ in} \end{aligned}$$

Condutividade térmica (k)

$$1 \text{ W}/(\text{m} \times \text{K}) = 0,577789 \text{ Btu}/(\text{h} \times \text{ft} \times \text{R}) \quad 1 \text{ Btu}/(\text{h} \times \text{ft} \times \text{R}) = 1,730735 \text{ W}/(\text{m} \times \text{K})$$

Constante universal dos gases

$$\begin{aligned} \bar{R} &= N_0 k = 8,31451 \text{ kJ}/\text{kmol} \times \text{K} & \bar{R} &= 1,98589 \text{ Btu}/\text{lbmol} \times \text{R} \\ &= 1,98589 \text{ kcal}/\text{kmol} \times \text{K} & &= 1545,36 \text{ lbg} \times \text{ft}/\text{lbmol} \times \text{R} \\ &= 82,0578 \text{ atm} \times \text{L}/\text{kmol} \times \text{K} & &= 0,73024 \text{ atm} \times \text{ft}^3/\text{lbmol} \times \text{R} \\ & & &= 10,7317 (\text{lbg}/\text{in}^2) \times \text{ft}^3/\text{lbmol} \times \text{R} \end{aligned}$$

Energia (E , U)

$$\begin{aligned} 1 \text{ J} &= 1 \text{ N} \times \text{m} = 0,737562 \text{ lbf} \times \text{ft} & 1 \text{ lbf} \times \text{ft} &= 1,355818 \text{ J} = 1,28507 \times 10^{-3} \text{ Btu} \\ 1 \text{ cal (Int.)} &= 4,1868 \text{ J} & 1 \text{ Btu (Int.)} &= 1,055056 \text{ kJ} = 778,1693 \text{ lbf} \times \text{ft} \end{aligned}$$

Energia cinética específica ($1/2 V^2$)

$$\begin{aligned} 1 \text{ m}^2/\text{s}^2 &= 0,001 \text{ kJ}/\text{kg} & 1 \text{ ft}^2/\text{s}^2 &= 3,9941 \times 10^{-5} \text{ Btu}/\text{lbm} \\ 1 \text{ kJ}/\text{kg} &= 1000 \text{ m}^2/\text{s}^2 & 1 \text{ Btu}/\text{lbm} &= 25037 \text{ ft}^2/\text{s}^2 \end{aligned}$$

Energia potencial específica (Zg)

$$\begin{aligned} 1 \text{ m} \times g_{\text{std}} &= 9,80665 \times 10^{-3} \text{ kJ}/\text{kg} & 1 \text{ ft} \times g_{\text{std}} &= 1,0 \text{ lbf}/\text{lbm} \\ &= 4,21607 \times 10^{-3} \text{ Btu}/\text{lbm} & &= 0,001285 \text{ Btu}/\text{lbm} \\ & & &= 0,002989 \text{ kJ}/\text{kg} \end{aligned}$$

Energia específica (e , u)

$$1 \text{ kJ}/\text{kg} = 0,42992 \text{ Btu}/\text{lbm} \quad 1 \text{ Btu}/\text{lbm} = 2,326 \text{ kJ}/\text{kg}$$

TABELA A.1Fatores de Conversão (*continuação*)

Fluxo de calor (por unidade de área)	
$1 \text{ W/m}^2 = 0,316998 \text{ Btu}/(\text{h} \times \text{ft}^2)$	$1 \text{ Btu}/(\text{h} \times \text{ft}^2) = 3,15459 \text{ W/m}^2$
Força (F)	
$1 \text{ N} = 0,224809 \text{ lbf}$ $1 \text{ kgf} = 9,80665 \text{ N}$	$1 \text{ lbf} = 4,448222 \text{ N}$
Gravidade	
$g = 9,80665 \text{ m/s}^2$	$g = 32,17405 \text{ ft/s}^2$
Massa (m)	
$1 \text{ kg} = 2,204623 \text{ lbm}$	$1 \text{ lbm} = 0,453592 \text{ kg}$
Massa específica (ρ)	
$1 \text{ kg/m}^3 = 0,06242797 \text{ lbm/ft}^3$	$1 \text{ lbm/ft}^3 = 16,01846 \text{ kg/m}^3$
Momento (Torque, T)	
$1 \text{ N} \times \text{m} = 0,737562 \text{ lbm} \times \text{ft}$	$1 \text{ lbm} \times \text{ft/s} = 1,355818 \text{ N} \times \text{m}$
Potência ($\dot{Q}$, $\dot{W}$)	
$1 \text{ W} = 1 \text{ J/s} = 0,737562 \text{ lbf} \times \text{ft/s}$ $1 \text{ kW} = 3412,14 \text{ Btu/h}$ $1 \text{ hp (métrico)} = 0,735499 \text{ kW}$ $1 \text{ tonelada de refrigeração} = 3,51685 \text{ kW}$	$1 \text{ lbf} \times \text{ft/s} = 1,355818 \text{ W} = 4,62624 \text{ Btu/h}$ $1 \text{ Btu/s} = 1,055056 \text{ kW}$ $1 \text{ hp (britânico)} = 0,7457 \text{ kW}$ $1 \text{ tonelada de refrigeração} = 12\,000 \text{ Btu/h}$
Pressão (P)	
$1 \text{ Pa} = 1 \text{ N/m}^2$ $1 \text{ bar} = 1,0 \times 10^5 \text{ Pa} = 100 \text{ kPa}$ $1 \text{ atm} = 101,325 \text{ kPa}$	$1 \text{ lbf} \times (\text{pol})^2 = 1 \text{ lbf} \times \text{in}^2 = 6,894757 \text{ kPa}$ $1 \text{ atm} = 14,69594 \text{ lbf} \times \text{in}^2$ $1 \text{ torr} = 1 \text{ mm Hg } [0^\circ \text{C}]$
Quantidade de movimento (mV)	
$1 \text{ kg} \times \text{m/s} = 7,23294 \text{ lbm} \times \text{ft/s}$ $= 0,224809 \text{ lbf} \times \text{s}$	$1 \text{ lbf} \times \text{ft} = 0,138256 \text{ kg} \times \text{m/s}$
Temperatura (T)	
$1 \text{ K} = 1^\circ \text{C}$ $\text{TC} = \text{TK} - 273,15$ $\text{TK} = \text{TR}/1,8$	$1 \text{ R} = (5/9) \text{ K}$ $\text{TF} = \text{TR} - 459,67$ $\text{TR} = 1,8 \text{ TK}$
Velocidade (V)	
$1 \text{ m/s} = 3,6 \text{ km/h}$ $= 3,28084 \text{ ft/s}$ $= 2,23694 \text{ mi/h}$ $1 \text{ km/h} = 0,27778 \text{ m/s}$ $= 0,91134 \text{ ft/s}$ $= 0,62137 \text{ mi/h}$	$1 \text{ ft/s} = 0,681818 \text{ mi/h}$ $= 0,3048 \text{ m/s}$ $= 1,09728 \text{ km/h}$ $1 \text{ mi/h} = 1,46667 \text{ ft/s}$ $= 0,44704 \text{ m/s}$ $= 1,609344 \text{ km/h}$
Volume (V)	
$1 \text{ m}^3 = 35,3147 \text{ ft}^3$ $1 \text{ litro} = 0,001 \text{ m}^3/\text{kg}$ $1 \text{ galão (britânico)} = 4,546090 \times 10^{-3} \text{ m}^3$	$1 \text{ ft}^3 = 2,831685 \times 10^{-2} \text{ m}^3$ $1 \text{ galão (US)} = 3,785412 \times 10^{-3} \text{ m}^3$
Volume específico (v)	
$1 \text{ cm}^3/\text{g} = 0,001 \text{ m}^3/\text{kg}$ $1 \text{ ft}^3/\text{lbm} = 0,062428 \text{ m}^3/\text{kg}$	$1 \text{ m}^3/\text{kg} = 16,01846 \text{ ft}^3/\text{lbm}$

TABELA A.2
Constantes críticas

Substância	Fórmula	Peso molecular	Temperatura K	Pressão MPa	Volume m ³ /kg
Amônia	NH ₃	17,031	405,5	11,35	0,00426
Argônio	Ar	39,948	150,8	4,87	0,00188
Bromo	Br ₂	159,808	588	10,30	0,000796
Dióxido de carbono	CO ₂	44,01	304,1	7,38	0,00212
Monóxido de carbono	CO	28,01	132,9	3,50	0,00333
Cloro	Cl ₂	70,906	416,9	7,98	0,00175
Flúor	F ₂	37,997	144,3	5,22	0,00174
Hélio	He	4,003	5,19	0,227	0,0143
Hidrogênio (normal)	H ₂	2,016	33,2	1,30	0,0323
Criptônio	Kr	83,80	209,4	5,50	0,00109
Neônio	Ne	20,183	44,4	2,76	0,00206
Óxido nítrico	NO	30,006	180	6,48	0,00192
Nitrogênio	N ₂	28,013	126,2	3,39	0,0032
Dióxido de nitrogênio	NO ₂	46,006	431	10,1	0,00365
Óxido nitroso	N ₂ O	44,013	309,6	7,24	0,00221
Oxigênio	O ₂	31,999	154,6	5,04	0,00229
Dióxido de enxofre	SO ₂	64,063	430,8	7,88	0,00191
Água	H ₂ O	18,015	647,3	22,12	0,00317
Xenônio	Xe	131,30	289,7	5,84	0,000902
Acetileno	C ₂ H ₂	26,038	308,3	6,14	0,00433
Benzeno	C ₆ H ₆	78,114	562,2	4,89	0,00332
<i>n</i> -Butano	C ₄ H ₁₀	58,124	425,2	3,80	0,00439
Difluorcloroetano ^a (142b)	CH ₃ CClF ₂	100,495	410,3	4,25	0,00230
Difluorclorometano (22)	CHClF ₂	86,469	369,3	4,97	0,00191
Diclorofluoreetano ^a (141)	CH ₃ CCl ₂ F	116,95	481,5	4,54	0,00215
Diclorotrifluoreetano ^a (123)	CHCl ₂ CF ₃	152,93	456,9	3,67	0,00182
Difluoreetano ^a (152a)	CHF ₂ CH ₃	66,05	386,4	4,52	0,00272
Difluormetano ^a (32)	CF ₂ H ₂	52,024	351,3	5,78	0,00236
Etano	C ₂ H ₆	30,070	305,4	4,88	0,00493
Álcool etílico (etanol)	C ₂ H ₅ OH	46,069	513,9	6,14	0,00363
Etileno	C ₂ H ₄	28,054	282,4	5,04	0,00465
<i>n</i> -Heptano	C ₇ H ₁₆	100,205	540,3	2,74	0,00431
<i>n</i> -Hexano	C ₆ H ₁₄	86,178	507,5	3,01	0,00429
Metano	CH ₄	16,043	190,4	4,60	0,00615
Álcool metílico (metanol)	CH ₃ OH	32,042	512,6	8,09	0,00368
<i>n</i> -Octano	C ₈ H ₁₈	114,232	568,8	2,49	0,00431
Pentafluoreetano (125)	CHF ₂ CF ₃	120,022	339,2	3,62	0,00176
<i>n</i> -Pentano	C ₅ H ₁₂	72,151	469,7	3,37	0,00421
Propano	C ₃ H ₈	44,094	369,8	4,25	0,00454
Propeno	C ₃ H ₆	42,081	364,9	4,60	0,00430
Refrigerante misto	R-410A	72,585	344,5	4,90	0,00218
Tetrafluoreetano (134a)	CF ₃ CH ₂ F	102,03	374,2	4,06	0,00197

TABELA A.3

Propriedades de alguns sólidos a 25 °C

Sólido	C_p , kJ/kg K	ρ , kg/m ³
Aço (AISI304)	0,46	7820
Acrílico	1,44	1180
Alumínio	0,9	2700
Areia (seca)	0,8	1500
Asfalto	0,92	2120
Borracha (macia)	1,67	1100
Carbono, diamante	0,51	3250
Carbono, grafite	0,61	2000–2500
Carvão	1,26	1200–1500
Chumbo	0,128	11340
Cobre	0,42	8300
Concreto	0,88	2200
Estanho	0,22	7304
Ferro fundido	0,42	7272
Gelo (0 °C)	2,04	917
Granito	0,89	2750
Lã	1,72	100
Lã de vidro	0,66	20
Latão, 60–40	0,38	8400
Madeira dura	1,26	720
Madeira mole (pinho)	1,38	510
Magnésio, 2% Mn	1,00	1778
Neve, firme	2,1	560
Níquel, 10% Cr	0,44	8666
Ouro	0,13	19300
Papel	1,2	700
Poliestireno	2,3	920
Prata, 99,9%	0,24	10524
PVC	0,96	1380
Sal, rocha	0,92	2100–2500
Silício	0,70	2330
Sódio	1,21	970
Tijolo (comum)	0,84	1800
Tungstênio	0,13	19300
Vidro	0,8	2500
Zinco	0,39	7144

TABELA A.4

Propriedades de alguns líquidos a 25 °C
(ou à temperatura de fusão, caso seja maior que 25 °C)

Líquido	C_p , kJ/kg K	ρ , kg/m ³
Água	4,184	997
Amônia	4,8	602
Benzeno	1,72	879
Butano	2,469	556
CCL ₄	0,83	1584
CO ₂	2,9	680
Estanho, Sn	0,24	6950
Etanol	2,46	783
Gasolina	2,08	750
Glicerina	2,42	1260
Metanol	2,55	787
<i>n</i> -octano	2,23	692
Óleo (leve)	1,8	910
Óleo de motor	1,9	885
Propano	2,54	510
Querosene	2,0	815
R-12	0,971	1310
R-125	1,41	1191
R-134a	1,43	1206
R-22	1,43	1206
R-32	1,94	961
R-410A	1,69	1059
Metais líquidos		
Bismuto, Bi	0,14	10040
Chumbo, Pb	0,16	10660
Mercúrio, Hg	0,14	13580
Na/K (56/44)	1,13	887
Potássio, K	0,81	828
Sódio, Na	1,38	929
Zinco, Zn	0,50	6570

TABELA A.5

Propriedades de vários gases ideais a 25 °C e 100 kPa*

Gás	Fórmula química	Peso molecular	R kJ/kg K	ρ (kg/m ³)	C_{p0} kJ/kg K	C_{v0} kJ/kg K	$k = C_p/C_v$
Acetileno	C ₂ H ₂	26,038	0,3193	1,05	1,699	1,380	1,231
Água (vapor)	H ₂ O	18,015	0,4615	0,0231	1,872	1,410	1,327
Ar		28,97	0,287	1,169	1,004	0,717	1,400
Amônia	NH ₃	17,031	0,4882	0,694	2,130	1,642	1,297
Argônio	Ar	39,948	0,2081	1,613	0,520	0,312	1,667
Butano	C ₄ H ₁₀	58,124	0,1430	2,407	1,716	1,573	1,091
Dióxido de Carbono	CO ₂	44,01	0,1889	1,775	0,842	0,653	1,289
Monóxido de Carbono	CO	28,01	0,2968	1,13	1,041	0,744	1,400
Etano	C ₂ H ₆	30,07	0,2765	1,222	1,766	1,490	1,186
Etanol	C ₂ H ₅ OH	46,069	0,1805	1,883	1,427	1,246	1,145
Etileno	C ₂ H ₄	28,054	0,2964	1,138	1,548	1,252	1,237
Hélio	He	4,003	2,0771	0,1615	5,193	3,116	1,667
Hidrogênio	H ₂	2,016	4,1243	0,0813	14,209	10,085	1,409
Metano	CH ₄	16,043	0,5183	0,648	2,254	1,736	1,299
Metanol	CH ₃ OH	32,042	0,2595	1,31	1,405	1,146	1,227
Neônio	Ne	20,183	0,4120	0,814	1,03	0,618	1,667
Nitrogênio	N ₂	28,013	0,2968	1,13	1,042	0,745	1,400
Óxido nítrico	NO	30,006	0,2771	1,21	0,993	0,716	1,387
Óxido nitroso	N ₂ O	44,013	0,1889	1,775	0,879	0,690	1,274
n-Octano	C ₈ H ₁₈	114,23	0,0727	0,092	1,7113	1,638	1,044
Oxigênio	O ₂	31,999	0,2598	1,292	0,9216	0,662	1,393
Propano	C ₃ H ₈	44,094	0,1886	1,808	1,679	1,490	1,126
R-12	CCl ₂ F ₂	120,914	0,06876	4,98	0,616	0,547	1,126
R-22	CHClF ₂	86,469	0,09616	3,54	0,658	0,562	1,171
R-32	CF ₂ H ₂	52,024	0,1598	2,125	0,822	0,662	1,242
R-125	CHF ₂ CF ₃	120,022	0,06927	4,918	0,791	0,722	1,097
R-134a	CF ₃ CH ₂ F	102,03	0,08149	4,20	0,852	0,771	1,106
Dióxido de enxofre	SO ₂	64,059	0,1298	2,618	0,624	0,494	1,263
Trióxido de enxofre	SO ₃	80,053	0,10386	3,272	0,635	0,531	1,196

* ou na pressão de saturação se esta for menor que 100 kPa

TABELA A.6

Calor específico a pressão constante de vários gases ideais em função da temperatura

		$C_{p0} = C_0 + C_1\theta + C_2\theta^2 + C_3\theta^3$ kJ/kg K				$\theta = T \text{ (Kelvin)}/1000$
Gás	Fórmula	C_0	C_1	C_2	C_3	Intervalo K
Acetileno	C_2H_2	1,03	2,91	-1,92	0,54	50-1200
Água (vapor)	H_2O	1,79	0,107	0,586	-0,20	50-1200
Amônia	NH_3	1,60	1,40	1,00	-0,7	50-1200
Ar		1,05	-0,365	0,85	-0,39	50-1200
Argônio	Ar	0,52	0	0	0	50-1200
Butano	C_4H_{10}	0,163	5,70	-1,906	-9,049	50-1200
Dióxido de carbono	CO_2	0,45	1,67	-1,27	0,39	250-1200
Dióxido de enxofre	SO_2	0,37	1,05	-0,77	0,21	250-1200
Etano	C_2H_6	0,18	5,92	-2,31	0,29	250-1200
Etanol	C_2H_5OH	0,20	-4,65	-1,82	0,03	250-1200
Etileno	C_2H_4	1,36	5,58	-3,00	0,63	250-1200
Hélio	He	5,193	0	0	0	250-1200
Hidrogênio	H_2	13,46	4,6	-6,85	3,79	250-1200
Metano	CH_4	1,2	3,25	0,75	-0,71	250-1200
Metanol	CH_3OH	0,66	2,21	0,81	-0,89	250-1200
Monóxido de carbono	CO	1,10	-0,46	1,00	-0,454	250-1200
Neônio	Ne	1,03	0	0	0	250-1200
Nitrogênio	N_2	1,11	-0,48	0,96	-0,42	250-1200
n-octano	C_8H_{18}	-0,053	6,75	-3,67	0,775	250-1200
Óxido nítrico	NO	0,98	-0,031	0,325	-0,14	250-1200
Óxido nitroso	N_2O	0,49	1,65	-1,31	0,42	250-1200
Oxigênio	O_2	0,88	-0,0001	0,54	-0,33	250-1200
Propano	C_3H_8	-0,096	6,95	-3,60	0,73	250-1200
R-12	CCl_2F_2	0,260	1,47	-1,25	0,36	250-500
R-22	$CHClF_2$	0,200	1,87	-1,35	0,35	250-500
R-32	CF_2H_2	0,227	2,27	-0,93	0,041	250-500
R-125	CHF_2CF_3	0,305	1,68	-0,284	0	250-500
R-134a	$C_2F_5CH_2F$	0,165	2,81	-2,23	1,11	250-500
Trióxido de enxofre	SO_3	0,24	1,70	-1,50	0,46	250-1200

As funções pressão relativa P_r e volume relativo v_r são dependentes da temperatura, calculadas com duas constantes A_1 e A_2 .

$$P_r = \exp\left[s_T^0/R - A_1\right]; \quad v_r = A_2 t/p_R$$

De modo que para um processo isoentrópico ($s_1 = s_2$):

$$\frac{P_2}{P_1} = \frac{P_{r2}}{P_{r1}} = \frac{e^{s_{T2}^0/R}}{e^{s_{T1}^0/R}} \approx \left(\frac{T_2}{T_1}\right)^{C_p/R} \quad e \quad \frac{v_2}{v_1} = \frac{v_{r2}}{v_{r1}} \approx \left(\frac{T_1}{T_2}\right)^{C_v/R}$$

Em que as aproximações se devem à hipótese de calor específico constante.

TABELA A.7

Propriedades termodinâmicas do ar (gás ideal; pressão de referência para a entropia é 0,1 MPa ou 1 bar)

T [K]	u kJ/kg	h kJ/kg	s_T^0 kJ/kg $\times$ K	P_r	v_r
200	142,77	200,17	6,46260	0,2703	493,47
220	157,07	220,22	6,55812	0,3770	389,15
240	171,38	240,27	6,64535	0,5109	313,27
260	185,70	260,32	6,72562	0,6757	256,58
280	200,02	280,39	6,79998	0,8756	213,26
290	207,02	290,43	6,83521	0,9899	195,36
298,15	213,04	298,62	6,86305	1,0907	182,29
300	214,36	300,47	6,86926	1,1146	179,49
320	228,73	320,58	6,93413	1,3972	152,73
340	243,11	340,70	6,99515	1,7281	131,20
360	257,53	360,86	7,05276	2,1123	113,65
380	271,99	381,06	7,10735	2,5548	99,188
400	286,49	401,30	7,15926	3,0612	87,137
420	301,04	421,59	7,20875	3,6373	77,003
440	315,64	441,94	7,25607	4,2897	68,409
460	330,31	462,34	7,30142	5,0233	61,066
480	345,04	482,81	7,34499	5,8466	54,748
500	359,84	503,36	7,38692	6,7663	49,278
520	374,73	523,98	7,42736	7,7900	44,514
540	389,69	544,69	7,46642	8,9257	40,344
560	404,74	565,47	7,50422	10,182	36,676
580	419,87	586,35	7,54084	11,568	33,436
600	435,10	607,32	7,57638	13,092	30,561
620	450,42	628,38	7,61090	14,766	28,001
640	465,83	649,53	7,64448	16,598	25,713
660	481,34	670,78	7,67717	18,600	23,662
680	496,94	692,12	7,70903	20,784	21,818
700	512,64	713,56	7,74010	23,160	20,155
720	528,44	735,10	7,77044	25,742	18,652
740	544,33	756,73	7,80008	28,542	17,289
760	560,32	778,46	7,82905	31,573	16,052
780	576,40	800,28	7,85740	34,851	14,925
800	592,58	822,20	7,88514	38,388	13,897
850	633,42	877,40	7,95207	48,468	11,695
900	674,82	933,15	8,01581	60,520	9,9170
950	716,76	989,44	8,07667	74,815	8,4677
1000	759,19	1046,22	8,13493	91,651	7,2760
1050	802,10	1103,48	8,19081	111,35	6,2885

TABELA A.7

Propriedades termodinâmicas do ar (*continuação*)

T [K]	u kJ/kg	h kJ/kg	s_T^0 kJ/kg $\times$ K	P_r	v_r
1100	845,45	1161,18	8,24449	134,25	5,4641
1150	881,21	1219,30	8,29616	160,73	4,7714
1200	933,37	1277,81	8,34596	191,17	4,1859
1250	977,89	1336,68	8,39402	226,02	3,6880
1300	1022,75	1395,89	8,44046	265,72	3,2626
1350	1067,94	1455,43	8,48539	310,74	2,8971
1400	1113,43	1515,27	8,52891	361,62	2,5817
1450	1159,20	1575,40	8,57111	418,89	2,3083
1500	1205,25	1635,80	8,61208	483,16	2,0703
1550	1251,55	1696,47	8,65185	554,96	1,8625
1600	1298,08	1757,33	8,69051	634,97	1,6804
1650	1344,83	1818,44	8,72811	723,86	1,52007
1700	1391,80	1879,76	8,76472	822,33	1,37858
1750	1438,97	1941,28	8,80039	931,14	1,25330
1800	1486,33	2002,99	8,83516	1051,05	1,14204
1850	1533,87	2064,88	8,86908	1182,9	1,04294
1900	1581,59	2126,95	8,90219	1327,5	0,95445
1950	1629,47	2189,19	8,93452	1485,8	0,87521
2000	1677,52	2251,58	8,96611	1658,6	0,80410
2050	1725,71	2314,13	8,99699	1847,1	0,74012
2100	1774,06	2376,82	9,02721	2052,1	0,68242
2150	1822,54	2439,66	9,05678	2274,8	0,63027
2200	1871,16	2502,63	9,08573	2516,2	0,58305
2250	1919,91	2565,73	9,11409	2777,5	0,54020
2300	1968,79	2628,96	9,14189	3059,9	0,50124
2350	2017,79	2692,31	9,16913	3364,6	0,46576
2400	2066,91	2755,78	9,19586	3693,0	0,43338
2450	2116,14	2819,37	9,22208	4046,2	0,40378
2500	2165,48	2883,06	9,24781	4425,8	0,37669
2550	2214,93	2946,86	9,27308	4833,0	0,35185
2600	2264,48	3010,76	9,29790	5269,5	0,32903
2650	2314,13	3074,77	9,32228	5736,7	0,30805
2700	2363,88	3138,87	9,34625	6236,2	0,28872
2750	2413,73	3203,06	9,36980	6769,7	0,27089
2800	2463,66	3267,35	9,39297	7338,7	0,25443
2850	2513,69	3331,73	9,41576	7945,1	0,23921
2900	2563,80	3396,19	9,43818	8590,7	0,22511
2950	2613,99	3460,73	9,46025	9277,2	0,21205
3000	2664,27	3525,36	9,48198	10007,0	0,19992

TABELA A.8

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base mássica

T (K)	Nitrogênio, Diatômico (N_2)			Oxigênio, Diatômico (O_2)		
	$R = 0,2968 \text{ kJ/kg} \times \text{K}$	$M = 28,013 \text{ kg/kmol}$		$R = 0,2598 \text{ kJ/kg} \times \text{K}$	$M = 31,999 \text{ kg/kmol}$	
	u (kJ/kg)	h (kJ/kg)	S_T^0 (kJ/kg $\times$ K)	u (kJ/kg)	h (kJ/kg)	S_T^0 (kJ/kg $\times$ K)
200	148,39	207,75	6,4250	129,84	181,81	6,0466
250	185,50	259,70	6,6568	162,41	227,37	6,2499
300	222,63	311,67	6,8463	195,20	273,15	6,4168
350	259,80	363,68	7,0067	228,37	319,31	6,5590
400	297,09	415,81	7,1459	262,10	366,03	6,6838
450	334,57	468,13	7,2692	296,52	413,45	6,7954
500	372,35	520,75	7,3800	331,72	461,63	6,8969
550	410,52	573,76	7,4811	367,70	510,61	6,9903
600	449,16	627,24	7,5741	404,46	560,36	7,0768
650	488,34	681,26	7,6606	441,97	610,86	7,1577
700	528,09	735,86	7,7415	480,18	662,06	7,2336
750	568,45	791,05	7,8176	519,02	713,90	7,3051
800	609,41	846,85	7,8897	558,46	766,33	7,3728
850	650,98	903,26	7,9581	598,44	819,30	7,4370
900	693,13	960,25	8,0232	638,90	872,75	7,4981
950	735,85	1017,81	8,0855	679,80	926,65	7,5564
1000	779,11	1075,91	8,1451	721,11	980,95	7,6121
1100	867,14	1193,62	8,2572	804,80	1090,62	7,7166
1200	957,00	1313,16	8,3612	889,72	1201,53	7,8131
1300	1048,46	1434,31	8,4582	975,72	1313,51	7,9027
1400	1141,35	1556,87	8,5490	1062,67	1426,44	7,9864
1500	1235,50	1680,70	8,6345	1150,48	1540,23	8,0649
1600	1330,72	1805,60	8,7151	1239,10	1654,83	8,1389
1700	1426,89	1931,45	8,7914	1328,49	1770,21	8,2088
1800	1523,90	2058,15	8,8638	1418,63	1886,33	8,2752
1900	1621,66	2185,58	8,9327	1509,50	2003,19	8,3384
2000	1720,07	2313,68	8,9984	1601,10	2120,77	8,3987
2100	1819,08	2442,36	9,0612	1693,41	2239,07	8,4564
2200	1918,62	2571,58	9,1213	1786,44	2358,08	8,5117
2300	2018,63	2701,28	9,1789	1880,17	2477,79	8,5650
2400	2119,08	2831,41	9,2343	1974,60	2598,20	8,6162
2500	2219,93	2961,93	9,2876	2069,71	2719,30	8,6656
2600	2321,13	3092,81	9,3389	2165,50	2841,07	8,7134
2700	2422,66	3224,03	9,3884	2261,94	2963,49	8,7596
2800	2524,50	3355,54	9,4363	2359,01	3086,55	8,8044
2900	2626,62	3487,34	9,4825	2546,70	3210,22	8,8478
3000	2729,00	3619,41	9,5273	2554,97	3334,48	8,8899

TABELA A.8 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base mássica

T (K)	Dióxido de carbono (CO_2)			Água (H_2O)		
	$R = 0,1889 \text{ kJ/kg} \times \text{K}$	$M = 44,010 \text{ kg/kmol}$		$R = 0,4615 \text{ kJ/kg} \times \text{K}$	$M = 18,015 \text{ kg/kmol}$	
	u (kJ/kg)	h (kJ/kg)	S_T^0 (kJ/kg $\times$ K)	u (kJ/kg)	h (kJ/kg)	S_T^0 (kJ/kg $\times$ K)
200	97,49	135,28	4,5439	276,38	368,69	9,7412
250	126,21	173,44	4,7139	345,98	461,36	10,1547
300	157,70	214,38	4,8631	415,87	554,32	10,4936
350	191,78	257,90	4,9972	486,37	647,90	10,7821
400	228,19	303,76	5,1196	557,79	742,40	11,0345
450	266,69	351,70	5,2325	630,40	838,09	11,2600
500	307,06	401,52	5,3375	704,36	935,12	11,4644
550	349,12	453,03	5,4356	779,79	1033,63	11,6522
600	392,72	506,07	5,5279	856,75	1133,67	11,8263
650	437,71	560,51	5,6151	935,31	1235,30	11,9890
700	483,97	616,22	5,6976	1015,49	1338,56	12,1421
750	531,40	673,09	5,7761	1097,35	1443,49	12,2868
800	579,89	731,02	5,8508	1180,90	1550,13	12,4244
850	629,35	789,93	5,9223	1266,19	1658,49	12,5558
900	676,69	849,72	5,9906	1353,23	1768,60	12,6817
950	730,85	910,33	6,0561	1442,03	1880,48	12,8026
1000	782,75	971,67	6,1190	1532,61	1994,13	12,9192
1100	888,55	1096,36	6,2379	1719,05	2226,73	13,1408
1200	996,64	1223,34	6,3483	1912,42	2466,25	13,3492
1300	1106,68	1352,28	6,4515	2112,47	2712,46	13,5462
1400	1218,38	1482,87	6,5483	2318,89	2965,03	13,7334
1500	1331,50	1614,88	6,6394	2531,28	3223,57	13,9117
1600	1445,85	1748,12	6,7254	2749,24	3487,69	14,0822
1700	1561,26	1882,43	6,8068	2972,35	3756,95	14,2454
1800	1677,61	2017,67	6,8841	3200,17	4030,92	14,4020
1900	1794,78	2153,73	6,9577	3432,28	4309,18	14,5524
2000	1912,67	2290,51	7,0278	3668,24	4591,30	14,6971
2100	2031,21	2427,95	7,0949	3908,08	4877,29	14,8366
2200	2150,34	2565,97	7,1591	4151,28	5166,64	14,9712
2300	2270,00	2704,52	7,2206	4397,56	5459,08	15,1012
2400	2390,14	2843,55	7,2798	4646,71	5754,37	15,2269
2500	2510,74	2983,04	7,3368	4898,49	6052,31	15,3485
2600	2631,73	3122,93	7,3917	5152,73	6352,70	15,4663
2700	2753,10	3263,19	7,4446	5409,24	6655,36	15,5805
2800	2874,81	3403,79	7,4957	5667,86	6960,13	15,6914
2900	2996,84	3544,71	7,5452	5928,44	7266,87	15,7990
3000	3119,18	3685,95	7,5931	6190,86	7575,44	15,9036

TABELA A.9

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base molar

T (K)	Nitrogênio diatômico (N_2) $\bar{h}_{f,298}^0 = 0 \text{ kJ/kmol}$ $M = 28,013 \text{ kg/kmol}$		Nitrogênio monatômico (N) $\bar{h}_{f,298}^0 = 472\,680 \text{ kJ/kmol}$ $M = 14,007 \text{ kg/kmol}$	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-8670	0	-6197	0
100	-5768	159,812	-4119	130,593
200	-2857	179,985	-2040	145,001
298	0	191,609	0	153,300
300	54	191,789	38	153,429
400	2971	200,181	2117	159,409
500	5911	206,740	4196	164,047
600	8894	212,177	6274	167,837
700	11937	216,865	8353	171,041
800	15046	221,016	10431	173,816
900	18223	224,757	12510	176,265
1000	21463	228,171	14589	178,455
1100	24760	231,314	16667	180,436
1200	28109	234,227	18746	182,244
1300	31503	236,943	20825	183,908
1400	34936	239,487	22903	185,448
1500	38405	241,881	24982	186,883
1600	41904	244,139	27060	188,224
1700	45430	246,276	29139	189,484
1800	48979	248,304	31218	190,672
1900	52549	250,234	33296	191,796
2000	56137	252,075	35375	192,863
2200	63362	255,518	39534	194,845
2400	70640	258,684	43695	196,655
2600	77963	261,615	47860	198,322
2800	85323	264,342	52033	199,868
3000	92715	266,892	56218	201,311
3200	100134	269,286	60420	202,667
3400	107577	271,542	64646	203,948
3600	115042	273,675	68902	205,164
3800	122526	275,698	73194	206,325
4000	130027	277,622	77532	207,437
4400	145078	281,209	86367	209,542
4800	160188	284,495	95457	211,519
5200	175352	287,530	104843	213,397
5600	190572	290,349	114550	215,195
6000	205848	292,984	124590	216,926

TABELA A.9 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base molar

T (K)	Oxigênio diatômico (O_2) $\bar{h}_{f,298}^0 = 0 \text{ kJ/kmol}$ $M = 21,999 \text{ kg/kmol}$		Oxigênio monatômico (O) $\bar{h}_{f,298}^0 = 249\,170 \text{ kJ/kmol}$ $M = 16,00 \text{ kg/kmol}$	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-8683	0	-6725	0
100	-5777	173,308	-4518	135,947
200	-2868	193,483	-2186	152,153
298	0	205,148	0	161,059
300	54	205,329	41	161,194
400	3027	213,873	2207	167,431
500	6086	220,693	4343	172,198
600	9245	226,450	6462	176,060
700	12499	231,465	8570	179,310
800	15836	235,920	10671	182,116
900	19241	239,931	12767	184,585
1000	22703	243,579	14860	186,790
1100	26212	246,923	16950	188,783
1200	29761	250,011	19039	190,600
1300	33345	252,878	21126	192,270
1400	36958	255,556	23212	193,816
1500	40600	258,068	25296	195,254
1600	44267	260,434	27381	196,599
1700	47959	262,673	29464	197,862
1800	51674	264,797	31547	199,053
1900	55414	266,819	33630	200,179
2000	59176	268,748	35713	201,247
2200	66770	272,366	39878	203,232
2400	74453	275,708	44045	205,045
2600	82225	278,818	48216	206,714
2800	90080	281,729	52391	208,262
3000	98013	284,466	56574	209,705
3200	106022	287,050	60767	211,058
3400	114101	289,499	64971	212,332
3600	122245	291,826	69190	213,538
3800	130447	294,043	73424	214,682
4000	138705	296,161	77675	215,773
4400	155374	300,133	86234	217,812
4800	172240	303,801	94873	219,691
5200	189312	307,217	103592	221,435
5600	206618	310,423	112391	223,066
6000	224210	313,457	121264	224,597

TABELA A.9 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a 0,1 × MPa) base molar

T (K)	Dióxido de carbono (CO ₂) $\bar{h}_{f,298}^0 = -393\,522$ kJ/kmol $M = 44,01$ kg/kmol		Monóxido de carbono (CO) $\bar{h}_{f,298}^0 = 110\,527$ kJ/kmol $M = 28,01$ kg/kmol	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-9364	0	-8671	0
100	-6457	179,010	-5772	165,852
200	-3413	199,976	-2860	186,024
298	0	213,794	0	197,651
300	69	214,024	54	197,831
400	4003	225,314	2977	206,240
500	8305	234,902	5932	212,833
600	12906	243,284	8942	218,321
700	17754	250,752	12021	223,067
800	22806	257,496	15174	227,277
900	28030	263,646	18397	231,074
1000	33397	269,299	21686	234,538
1100	38885	274,528	25031	237,726
1200	44473	279,390	28427	240,679
1300	50148	283,931	31867	243,431
1400	55895	288,190	35343	246,006
1500	61705	292,199	38852	248,426
1600	67569	295,984	42388	250,707
1700	73480	299,567	45948	252,866
1800	79432	302,969	49529	254,913
1900	85420	306,207	53128	256,860
2000	91439	309,294	56743	258,716
2200	103562	315,070	64012	262,182
2400	115779	320,384	71326	265,361
2600	128074	325,307	78679	268,302
2800	140435	329,887	86070	271,044
3000	152853	334,170	93504	273,607
3200	165321	338,194	100962	276,012
3400	177836	341,988	108440	278,279
3600	190394	345,576	115938	280,422
3800	202990	348,981	123454	282,454
4000	215624	352,221	130989	284,387
4400	240992	358,266	146108	287,989
4800	266488	363,812	161285	291,290
5200	292112	368,939	176510	294,337
5600	317870	373,711	191782	297,167
6000	343782	378,180	207105	299,809

TABELA A.9 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a 0,1 × MPa) base molar

T (K)	Água (H ₂ O) $\bar{h}_{f,298}^0 = -241\,826$ kJ/kmol $M = 18,015$ kg/kmol		Hidroxila (OH) $\bar{h}_{f,298}^0 = 38\,987$ kJ/kmol $M = 17,007$ kg/kmol	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-9904	0	-9172	0
100	-6617	152,386	-6140	149,591
200	-3282	175,488	-2975	171,592
298	0	188,835	0	183,709
300	62	189,043	55	183,894
400	3450	198,787	3034	192,466
500	6922	206,532	5991	199,066
600	10499	213,051	8943	204,448
700	14190	218,739	11902	209,008
800	18002	223,826	14881	212,984
900	21937	228,460	17889	216,526
1000	26000	232,739	20935	219,735
1100	30190	236,732	24024	222,680
1200	34506	240,485	27159	225,408
1300	38941	244,035	30340	227,955
1400	43491	247,406	33567	230,347
1500	48149	250,620	36838	232,604
1600	52907	253,690	40151	234,741
1700	57757	256,631	43502	236,772
1800	62693	259,452	46890	238,707
1900	67706	262,162	50311	240,556
2000	72788	264,769	53763	242,328
2200	83153	269,706	60751	245,659
2400	93741	274,312	67840	248,743
2600	104520	278,625	75018	251,614
2800	115463	282,680	82268	254,301
3000	126548	286,504	89585	256,825
3200	137756	290,120	96960	259,205
3400	149073	293,550	104388	261,456
3600	160484	296,812	111864	263,592
3800	171981	299,919	119382	265,625
4000	183552	302,887	126940	267,563
4400	206892	308,448	142165	271,191
4800	230456	313,573	157522	274,531
5200	254216	318,328	173002	277,629
5600	278161	322,764	188598	280,518
6000	302295	326,926	204309	283,227

TABELA A.9 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base molar

T (K)	Hidrogênio (H_2) $\bar{h}_{f,298}^0 = 0 \text{ kJ/kmol}$ $M = 2,016 \text{ kg/kmol}$		Hidrogênio monatômico (H) $\bar{h}_{f,298}^0 = 217\,999 \text{ kJ/kmol}$ $M = 1,008 \text{ kg/kmol}$	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-8467	0	-6197	0
100	-5467	100,727	-4119	92,009
200	-2774	119,410	-2040	106,417
298	0	130,678	0	114,716
300	53	130,856	38	114,845
400	2961	139,219	2117	120,825
500	5883	145,738	4196	125,463
600	8799	151,078	6274	129,253
700	11730	155,609	8353	132,457
800	14681	159,554	10431	135,233
900	17657	163,060	12510	137,681
1000	20663	166,225	14589	139,871
1100	23704	169,121	16667	141,852
1200	26785	171,798	18746	143,661
1300	29907	174,294	20825	145,324
1400	33073	176,637	22903	146,865
1500	36281	178,849	24982	148,299
1600	39533	180,946	27060	149,640
1700	42826	182,941	29139	150,900
1800	46160	184,846	31218	152,089
1900	49532	186,670	33296	153,212
2000	52942	188,419	35375	154,279
2200	59865	191,719	39532	156,260
2400	66915	194,789	43689	158,069
2600	74082	197,659	47847	159,732
2800	81355	200,355	52004	161,273
3000	88725	202,898	56161	162,707
3200	96187	205,306	60318	164,048
3400	103736	207,593	64475	165,308
3600	111367	209,773	68633	166,497
3800	119077	211,856	72790	167,620
4000	126864	213,851	76947	168,687
4400	142658	217,612	85261	170,668
4800	158730	221,109	93576	172,476
5200	175057	224,379	101890	174,140
5600	191607	227,447	110205	175,681
6000	208332	230,322	118519	177,114

TABELA A.9 (continuação)

Propriedades de várias substâncias (gases ideais, entropias a $0,1 \times \text{MPa}$) base molar

T (K)	Óxido nítrico (NO) $\bar{h}_{f,298}^0 = 90\,291 \text{ kJ/kmol}$ $M = 30,006 \text{ kg/kmol}$		Dióxido de nitrogênio (NO_2) $\bar{h}_{f,298}^0 = 33\,100 \text{ kJ/kmol}$ $M = 46,005 \text{ kg/kmol}$	
	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K	$(\bar{h} - \bar{h}_{f,298}^0)$ kJ/kmol	$\bar{s}_T^0$ kJ/kmol K
0	-9192	0	-10186	0
100	-6073	177,031	-6861	202,563
200	-2951	198,747	-3495	225,852
298	0	210,759	0	240,034
300	55	210,943	68	240,263
400	3040	219,529	3927	251,342
500	6059	226,263	8099	260,638
600	9144	231,886	12555	268,755
700	12308	236,762	17250	275,988
800	15548	241,088	22138	282,513
900	18858	244,985	27180	288,450
1000	22229	248,536	32344	293,889
1100	25653	251,799	37606	298,904
1200	29120	254,816	42946	303,551
1300	32626	257,621	48351	307,876
1400	36164	260,243	53808	311,920
1500	39729	262,703	59309	315,715
1600	43319	265,019	64846	319,289
1700	46929	267,208	70414	322,664
1800	50557	269,282	76008	325,861
1900	54201	271,252	81624	328,898
2000	57859	273,128	87259	331,788
2200	65212	276,632	98578	337,182
2400	72606	279,849	109948	342,128
2600	80034	282,822	121358	346,695
2800	87491	285,585	132800	350,934
3000	94973	288,165	144267	354,890
3200	102477	290,587	155756	358,597
3400	110000	292,867	167262	362,085
3600	117541	295,022	178783	365,378
3800	125099	297,065	190316	368,495
4000	132671	299,007	201860	371,456
4400	147857	302,626	224973	376,963
4800	163094	305,940	248114	381,997
5200	178377	308,998	271276	386,632
5600	193703	311,838	294455	390,926
6000	209070	314,488	317648	394,926

TABELA A.10

Entalpia de formação e entropia absoluta de várias substâncias a 25°C e 100 kPa

Substância	Fórmula	M	Estado	$\bar{h}_f^0$ kJ/kmol	$\bar{s}_f^0$ kJ/kmol K
Acetileno	C_2H_2	26,038	gás	+226 731	200,958
Água	H_2O	18,015	gás	-241 826	188,834
Água	H_2O	18,015	líquido	-285 830	69,950
Amônia	NH_3	17,031	gás	-45 720	192,572
Benzeno	C_6H_6	78,114	gás	+82 980	269,562
Butano	C_4H_{10}	58,124	gás	-126 200	306,647
Carbono (grafita)	C	12,011	sólido	0	5,740
Dióxido de carbono	CO_2	44,010	gás	-393 522	213,795
Dióxido de enxofre	SO_2	64,059	gás	-296 842	248,212
Enxofre	S	32,06	sólido	0	32,056
Etano	C_2H_6	30,070	gás	-84 740	229,597
Etanol	C_2H_5OH	46,069	gás	-235 000	282,444
Etanol	C_2H_5OH	46,069	líquido	-277 380	160,554
Eteno	C_2H_4	28,054	gás	+52 467	219,330
Heptano	C_7H_{16}	100,205	gás	-187 900	427,805
Hexano	C_6H_{14}	86,178	gás	-167 300	387,979
Metano	CH_4	16,043	gás	-74 873	186,251
Metanol	CH_3OH	32,042	gás	-201 300	239,709
Metanol	CH_3OH	32,042	líquido	-239 220	126,809
Monóxido de carbono	CO	28,011	gás	-110 527	197,653
Nitrometano	CH_3NO_2	61,04	líquido	-113 100	171,80
<i>n</i> -Octano	C_8H_{18}	114,232	gás	-208 600	466,514
<i>n</i> -Octano	C_8H_{18}	114,232	líquido	-250 105	360,575
Óxido de Nitrogênio	N_2O	44,013	gás	+82 050	219,957
Ozônio	O_3	47,998	gás	+142 674	238,932
Pentano	C_5H_{12}	72,151	gás	-146 500	348,945
Peróxido de hidrogênio	H_2O_2	34,015	gás	-136 106	232,991
Propano	C_3H_8	44,094	gás	-103 900	269,917
Propeno	C_3H_6	42,081	gás	+20 430	267,066
Trióxido de enxofre	SO_3	80,058	gás	-395 765	256,769
<i>T</i> -Diesel	$C_{14,4}H_{24,9}$	198,06	líquido	-174 000	525,90

TABELA A.11

Logaritmos na base e da constante de equilíbrio K

Para a reação $\nu_A A + \nu_B B \rightleftharpoons \nu_C C + \nu_D D$, a constante de equilíbrio K é definida por:

Temp K	$H_2 \rightleftharpoons 2H$	$O_2 \rightleftharpoons 2O$	$N_2 \rightleftharpoons 2N$	$2H_2O \rightleftharpoons 2H_2 + O_2$	$2H_2O \rightleftharpoons H_2 + 2OH$	$2CO_2 \rightleftharpoons 2CO + O_2$	$N_2 + O_2 \rightleftharpoons 2NO$	$N_2 + 2O_2 \rightleftharpoons 2NO_2$
298	-164,003	-186,963	-367,528	-184,420	-212,075	-207,529	-69,868	-41,355
500	-92,830	-105,623	-213,405	-105,385	-120,331	-115,234	-40,449	-30,725
1000	-39,810	-45,146	-99,146	-46,321	-51,951	-47,052	-18,709	-23,039
1200	-30,878	-35,003	-80,025	-36,363	-40,467	-35,736	-15,082	-21,752
1400	-24,467	-27,741	-66,345	-29,222	-32,244	-27,679	-12,491	-20,826
1600	-19,638	-22,282	-56,069	-23,849	-26,067	-21,656	-10,547	-20,126
1800	-15,868	-18,028	-48,066	-19,658	-21,258	-16,987	-9,035	-19,577
2000	-12,841	-14,619	-41,655	-16,299	-17,406	-13,266	-7,825	-19,136
2200	-10,356	-11,826	-36,404	-13,546	-14,253	-10,232	-6,836	-18,773
2400	-8,280	-9,495	-32,023	-11,249	-11,625	-7,715	-6,012	-18,470
2600	-6,519	-7,520	-28,313	-9,303	-9,402	-5,594	-5,316	-18,214
2800	-5,005	-5,826	-25,129	-7,633	-7,496	-3,781	-4,720	-17,994
3000	-3,690	-4,356	-22,367	-6,184	-5,845	-2,217	-4,205	-17,805
3200	-2,538	-3,069	-19,947	-4,916	-4,401	-0,853	-3,755	-17,640
3400	-1,519	-1,932	-17,810	-3,795	-3,128	0,346	-3,359	-17,496
3600	-0,611	-0,922	-15,909	-2,799	-1,996	1,408	-3,008	-17,369
3800	0,201	-0,017	-14,205	-1,906	-0,984	2,355	-2,694	-17,257
4000	0,934	0,798	-12,671	-1,101	-0,074	3,204	-2,413	-17,157
4500	2,483	2,520	-9,423	0,602	1,847	4,985	-1,824	-16,953
5000	3,724	3,898	-6,816	1,972	3,383	6,397	-1,358	-16,797
5500	4,739	5,027	-4,672	3,098	4,639	7,542	-0,980	-16,678
6000	5,587	5,969	-2,876	4,040	5,684	8,488	-0,671	-16,588

Fonte: Consistente com JANAF Thermochemical Tables, 3ª edição, Thermal Group, Dow Chemical U.S.A., Mid., MI, 1985.

TABELA A.12

Funções de escoamento compressível para escoamento isentrópico unidimensional (gás ideal com calor específico e massa molecular constantes; $k = 1,4$).

M	M^*	A/A^*	P/P_0	ρ/ρ_0	T/T_0
0,0	0,00000	∞	1,00000	1,00000	1,00000
0,1	0,10944	5,82183	0,99303	0,99502	0,99800
0,2	0,21822	2,96352	0,97250	0,98028	0,99206
0,3	0,32572	2,03506	0,93947	0,95638	0,98232
0,4	0,43133	1,59014	0,89561	0,92427	0,96899
0,5	0,53452	1,33984	0,84302	0,88517	0,95238
0,6	0,63481	1,18820	0,78400	0,84045	0,93284
0,7	0,73179	1,09437	0,72093	0,79158	0,91075
0,8	0,82514	1,03823	0,65602	0,73999	0,88652
0,9	0,91460	1,00886	0,59126	0,68704	0,86059
1,0	1,0000	1,00000	0,52828	0,63394	0,83333
1,1	1,0812	1,00793	0,46835	0,58170	0,80515
1,2	1,1583	1,03044	0,41238	0,53114	0,77640
1,3	1,2311	1,06630	0,36091	0,48290	0,74738
1,4	1,2999	1,11493	0,31424	0,43742	0,71839
1,5	1,3646	1,17617	0,27240	0,39498	0,68966
1,6	1,4254	1,25023	0,23527	0,35573	0,66138
1,7	1,4825	1,33761	0,20259	0,31969	0,63371
1,8	1,5360	1,43898	0,17404	0,28682	0,60680
1,9	1,5861	1,55526	0,14924	0,25699	0,58072
2,0	1,6330	1,68750	0,12780	0,23005	0,55556
2,1	1,6769	1,83694	0,10925	0,20580	0,53135
2,2	1,7179	2,00497	0,93422E-01	0,18405	0,50813
2,3	1,7563	2,19313	0,79973E-01	0,16458	0,48591
2,4	1,7922	2,40310	0,68399E-01	0,14720	0,46468
2,5	1,8257	2,63672	0,58528E-01	0,13169	0,44444
2,6	1,8571	2,89598	0,50115E-01	0,11787	0,42517
2,7	1,8865	3,18301	0,42950E-01	0,10557	0,40683
2,8	1,9140	3,50012	0,36848E-01	0,94626E-01	0,38941
2,9	1,9398	3,84977	0,31651E-01	0,84889E-01	0,37286
3,0	1,9640	4,23457	0,27224E-01	0,76226E-01	0,35714
3,5	2,0642	6,78962	0,13111E-01	0,45233E-01	0,28986
4,0	2,1381	10,7188	0,65861E-02	0,27662E-01	0,23810
4,5	2,1936	16,5622	0,34553E-02	0,17449E-01	0,19802
5,0	2,2361	25,0000	0,18900E-02	0,11340E-01	0,16667
6,0	2,2953	53,1798	0,63336E-03	0,51936E-02	0,12195
7,0	2,3333	104,143	0,24156E-03	0,26088E-02	0,09259
8,0	2,3591	190,109	0,10243E-03	0,14135E-02	0,07246
9,0	2,3772	327,198	0,47386E-04	0,81504E-03	0,05814
10,0	2,3905	535,938	0,23563E-04	0,49482E-03	0,04762
∞	2,4495	∞	0,0	0,0	0,0

TABELA A.13

Funções de choque normal para escoamento compressível unidimensional (gás ideal com calor específico e massa molecular constantes; $k = 1,4$).

M_x	M_y	P_y/P_x	ρ_y/ρ_x	T_y/T_x	P_{0y}/P_{0x}	P_{0y}/P_x
1,00	1,00000	1,0000	1,0000	1,0000	1,00000	1,8929
1,05	0,95313	1,1196	1,0840	1,0328	0,99985	2,0083
1,10	0,91177	1,2450	1,1691	1,0649	0,99893	2,1328
1,15	0,87502	1,3763	1,2550	1,0966	0,99669	2,2661
1,20	0,84217	1,5133	1,3416	1,1280	0,99280	2,4075
1,25	0,81264	1,6563	1,4286	1,1594	0,98706	2,5568
1,30	0,78596	1,8050	1,5157	1,1909	0,97937	2,7136
1,35	0,76175	1,9596	1,6028	1,2226	0,96974	2,8778
1,40	0,73971	2,1200	1,6897	1,2547	0,95819	3,0492
1,45	0,71956	2,2863	1,7761	1,2872	0,94484	3,2278
1,50	0,70109	2,4583	1,8621	1,3202	0,92979	3,4133
1,55	0,68410	2,6362	1,9472	1,3538	0,91319	3,6057
1,60	0,66844	2,8200	2,0317	1,3880	0,89520	3,8050
1,65	0,65396	3,0096	2,1152	1,4228	0,87599	4,0110
1,70	0,64054	3,2050	2,1977	1,4583	0,85572	4,2238
1,75	0,62809	3,4063	2,2791	1,4946	0,83457	4,4433
1,80	0,61650	3,6133	2,3592	1,5316	0,81268	4,6695
1,85	0,60570	3,8263	2,4381	1,5693	0,79023	4,9023
1,90	0,59562	4,0450	2,5157	1,6079	0,76736	5,1418
1,95	0,58618	4,2696	2,5919	1,6473	0,74420	5,3878
2,00	0,57735	4,5000	2,6667	1,6875	0,72087	5,6404
2,05	0,56906	4,7362	2,7400	1,7285	0,69751	5,8996
2,10	0,56128	4,9783	2,8119	1,7705	0,67420	6,1654
2,15	0,55395	5,2263	2,8823	1,8132	0,54105	6,4377
2,20	0,54706	5,4800	2,9512	1,8569	0,62814	6,7165
2,25	0,54055	5,7396	3,0186	1,9014	0,60553	7,0018
2,30	0,53441	6,0050	3,0845	1,9468	0,58329	7,2937
2,35	0,52861	6,2762	3,1490	1,9931	0,56148	7,5920
2,40	0,52312	6,5533	3,2119	2,0403	0,54014	7,8969
2,45	0,51792	6,8363	3,2733	2,0885	0,51931	8,2083
2,50	0,51299	7,1250	3,3333	2,1375	0,49901	8,5261
2,55	0,50831	7,4196	3,3919	2,1875	0,47928	8,8505
2,60	0,50387	7,7200	3,4490	2,2383	0,46012	9,1813
2,70	0,49563	8,3383	3,5590	2,3429	0,42359	9,8624
2,80	0,48817	8,9800	3,6636	2,4512	0,38946	10,569
2,90	0,48138	9,6450	3,7629	2,5632	0,35773	11,302
3,00	0,47519	10,333	3,8571	2,6790	0,32834	12,061
4,00	0,43496	18,500	4,5714	4,0469	0,13876	21,068
5,00	0,41523	29,000	5,0000	5,8000	0,06172	32,653
10,00	0,38758	116,5	5,7143	20,387	0,00304	129,22